

Education

Seoul National University	Mar 2020 – Feb 2027 (Expected)
Ph.D. in Computer Science and Engineering – Human-Centered Computer Systems Lab	
Advised by Youngki Lee	Seoul, Republic of Korea
Seoul National University	Mar 2018 – Feb 2020
M.S. in Computer Science and Engineering – Computer Graphics & Image Processing Lab	
Advised by Yeong-Gil Shin	Seoul, Republic of Korea
DigiPen Institute of Technology	Mar 2012 – Apr 2016
B.S. in Computer Science in Real-Time Interactive Simulation	Redmond, Washington, USA
DigiPen Institute of Technology Europe	Sep 2013 – Dec 2013
Exchange Semester	Bilbao, Spain

Work Experience

Microsoft Research Asia	Dec 2020 – May 2021
Research Intern — Heterogeneous and Extreme Computing (HEX) Group	
Advised by Yunxin Liu and Ting Cao	Beijing, China (Virtual)
Microsoft	Sep 2016 – May 2017
Software Engineer — Minecraft: Education Edition Team	Redmond, Washington, USA

Publication ([Google Scholar](#), †: Equal Contribution)

How Physics Fidelity Shapes Perceived Plausibility in Mixed Reality: A Case Study of Bounce Behavior	Oct 2026
Hyunsoo Kim, Jingyu Lee , Rajesh Balan, Youngki Lee	
IEEE Intl. Symposium on Mixed and Augmented Reality (ISMAR)	
Probing Ambient Co-Presence as Social Infrastructure: First Encounters with Connected Rooms	June 2026
Jingyu Lee , Youngki Lee	
ACM Intl. Workshop on Social Infrastructure Systems (SocialSys, with BuildSys) [DOI]	
Logan: Loss-tolerant Live Video Analytics System	Nov 2024
Kichang Yang, Minkyung Jeong, Juheon Yi, Jingyu Lee , Kyungsoo Park, Youngki Lee	
ACM Intl. Conference on Mobile Computing and Networking (MobiCom) [DOI]	
Maestro: The Analysis-Simulation Integrated Framework for Mixed Reality	June 2024
Jingyu Lee , Hyunsoo Kim, Minjae Kim, Byung-Gon Chun, Youngki Lee	
ACM Intl. Conference on Mobile Systems, Applications, and Services (MobiSys) [DOI]	

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors June 2022
Joo Seong Jeong[†], **Jingyu Lee**[†], Donghyun Kim, Changmin Jeon, Changjin Jeong, Youngki Lee, Byung-Gon Chun

ACM Intl. Conference on Mobile Systems, Applications, and Services (MobiSys) [DOI]

ParallelFusion: Towards Maximum Utilization of Mobile GPU for DNN Inference June 2021
Jingyu Lee, Yunxin Liu, Youngki Lee

ACM Intl. Workshop on Embedded and Mobile Deep Learning (EMDL, with MobiSys) [DOI]

Liver Segmentation in Abdominal CT Images via Auto-Context Neural Network and Self-supervised Contour Attention Mar 2021

Minyoung Chung, **Jingyu Lee**, Jeongjin Lee, Yeong-Gil Shin

Artificial Intelligence in Medicine [DOI]

Automatic Registration between Cone-Beam CT and Scanned Surface via Deep-Pose Regression Neural Networks and Clustered Similarities Dec 2020

Minyoung Chung, **Jingyu Lee**, Wisoo Song, Youngchan Song, Il-Hyung Yang, Jeongjin Lee, Yeong-Gil Shin

IEEE Transactions on Medical Imaging [DOI]

Deeply Self-Supervised Contour Embedded Neural Network Applied to Liver Segmentation Aug 2020

Minyoung Chung, **Jingyu Lee**, Minkyung Lee, Jeongjin Lee, Yeong-Gil Shin

Computer Methods and Programs in Biomedicine [DOI]

Pose-Aware Instance Segmentation Framework from Cone Beam CT Images for Tooth Segmentation May 2020

Minyoung Chung, Minkyung Lee, Jioh Hong, Sanguk Park, Jusang Lee, **Jingyu Lee**, Il-Hyung Yang, Jeongjin Lee, Yeong-Gil Shin

Computers in Biology and Medicine [DOI]

Publication (Preprint / Under Review)

SpatialBlend: Relational Reconstruction for Cross-Space Presence in Heterogeneous Spaces

Jingyu Lee, Dongho Han, Jinmyeong Kim, Youngki Lee

Under Review

Explorable Ideas: Externalizing Ideas as Explorable Environments Dec 2025

Euijun Jung, **Jingyu Lee**, Minji Kim, Youngki Lee

arXiv:2512.17017 [DOI]

Project

Band: Multi-DNN Framework for Mobile Platform

Project Lead & Main Developer [website] [github] [UE plugin] [examples]

Somnus (Senior year student project at DigiPen)

Technical Director (Team of 4 Programmers, 1 Designer, 4 Artists, 3 Sound Designers)

3D Puzzle Platformer using perspective illusions [link]

Honors & Award

E3 College Game Competition — Somnus	Jun 2016
Showcased at Los Angeles Convention Center [DigiPen news] [Interview]	Finalist (top 5 out of ~400)
Bisa Scholarship	Mar 2012
Full-ride scholarship granted from DigiPen - KMU dual degree program	~\$60,000

Teaching

SNU & LG Data Science Course: Machine Learning	Jan 2021 - 2024, Feb 2020
Machine Learning Theory and Practice	Teaching Assistant
SNU & Samsung Data Science Course: Machine Learning	Oct 2023, Jun 2023
Machine Learning Projects with Kaggle	Teaching Assistant
SNU & SK Hynix Machine Learning Engineer Course	Nov 2022, Aug 2022
Time Series Data Analysis	Teaching Assistant
SNU Big Data & Fintech Expert Training Course	Apr 2021
Mobile Deep Learning & Machine Learning Application Development	Teaching Assistant
SNU Computer Programming (M1522.000600, 001)	Fall 2020
Object Oriented Programming in Java & C++	Teaching Assistant

Mentoring

Heeseo Kim	Jul 2025 – Feb 2026
Undergrad Research Intern, Human-Centered Computer Systems Lab	
XR Agents	
Junhee Kim	Jul 2025 – Present
Undergrad Research Intern, Human-Centered Computer Systems Lab	
XR Agents	
Euijoon Jung	Feb 2025 – Sep 2025
Ph.D. Student, Human-Centered Computer Systems Lab	
Idea Island (XR Ideation Tool)	
Jinmyeong Kim	Jun 2024 – Present
Ph.D. Student, Human-Centered Computer Systems Lab	
SpatialBlend ; XR Perception Platform (On-device Hand-Object Perception)	
Dongho Han	Jun 2024 – Present
Ph.D. Student, Human-Centered Computer Systems Lab	
SpatialBlend ; XR Perception Platform (On-device Hand-Object Perception)	

Hyunsoo Kim

Mar 2022 – Present

Undergrad Research Intern → Ph.D. Student, Human-Centered Computer Systems Lab

Maestro; MR Physics Fidelity

Invited Talk

University of Texas at Austin

Dec 2024

Analysis-Simulation Integrated Platform For Mixed Reality

Host: Daehyeok Kim

Duke University

Dec 2024

Analysis-Simulation Integrated Platform For Mixed Reality

Host: Maria Gorlatova

University of California, Merced

Dec 2024

Analysis-Simulation Integrated Platform For Mixed Reality

Host: Shijia Pan

KAIST School of Computing

Sep 2023

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors

Host: Junehwa Song

Samsung Global Technology Symposium — Poster

Apr 2023

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors

Institute of Embedded Engineering of Korea Fall Conference Proceedings

Nov 2022

Band: Coordinated Multi-DNN Inference on Heterogeneous Mobile Processors

A3 Foresight Workshop

Aug 2021

ParallelFusion: Towards Maximum Utilization of Mobile GPU for DNN Inference [\[link\]](#)

Academic Service

IEEE Transactions on Vehicular Technology

Reviewer

IEEE Transactions on Medical Imaging

Reviewer

updated: June 25, 2026